



LCI Advanced Workshop 2025: Licenses

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Licenses Overview

- Introduction and Objectives
 - What is the goal?
 - Who will use these?
 - Why use licenses?
- Fundamentals
 - Syntax for Allocating
 - Local Licenses vs Remote Licenses
 - Trackable Resources and Limits

What is the Goal?

Things Admins Care About:

- Site license compliance
- Usage auditability
- Hands-off integration

Things Users Care About:

- Being able to open Ansys Fluent
- ...?

Who Uses Licenses?

Licenses should be attractive to any site where resources are freely usable on nodes, but there needs to be a clean way to limit simultaneous usage.

A site with infinite Matlab seats will not want to implement Slurm-based licenses.

Many licensed-seat software are straightforward enough to work with finite availability that licenses still may be foregone.

Why Use Licenses?

Licenses can help provide lessened friction on interactive and batched jobs depending on software with limited simultaneous use.

A worst-case scenario is a batched job/job array that somehow uses an amount of licenses greater than those available, queuing up multiple failures of the job.

Syntax for Allocating

For interactive allocations:

```
$ salloc -L matlab:1
```

```
$ salloc -L matlab:1,a100_40:1
```

```
$ salloc --licenses=matlab25|matlab24
```

For batched allocations:

```
#SBATCH --licenses=matlab:1
```

```
#SBATCH -L matlab:4
```

Licenses can be requested through the short or long-form, which operate identically.

, (commas) between licenses signify **AND** – both licenses requested and required for the job to start.

| (pipes) between licenses signify **OR** – either license required for job to start.

License Types

Local Licenses

- Local to the cluster
- Easiest Accounting
- Just works

A license where ``Licenses=fluent:50`` simply means “50 concurrent jobs may checkout this license.”

This means you can readily apply licenses to:

- Software seats
- Arbitrary hardware, e.g., FPGAs, GPUs
- An entire node

Local licenses can be viewed as **cluster-wide consumable resources**, rather than **node-specific**.

License Types

Remote Licenses

- Means the licensing is tracked by the database
- Is **not** an integration with third party license managers
- Dynamically updateable
- Shareable across multiple clusters

The reason these exist are so that you can attribute license allocation across multiple clusters.

```
sacctmgr add resource name=matlab server=rlm_host type=license count=50
sacctmgr add resource name=matlab server=rlm_host cluster=oldone allowed=30
sacctmgr add resource name=matlab server=rlm_host cluster=newone allowed=70
```

Notice anything peculiar about the math?

License Types

Remote Licenses

```
sacctmgr add resource name=matlab server=rlm_host type=license count=50  
sacctmgr add resource name=matlab server=rlm_host cluster=oldone allowed=70  
sacctmgr add resource name=matlab server=rlm_host cluster=newone allowed=30
```

One thing to note is that we are declaring the limits of each cluster, **oldone** and **newone** to have 30 and 70 licenses simultaneously.

But why, if there are only 50 licenses?

Because licenses are *dynamic*, and you might have 50 permitted now, but 150 another time, or expecting to in the future. Regardless, if you know now that you never want there to be an allocation of > allowed number on a cluster, you can specify it, simply to be precise.

Whether we foresee our own site to have this need, Slurm builds in the capability to put a ceiling on a per-cluster basis.

License Types

Remote Licenses “server”

```
sacctmgr add resource name=matlab server=rlm_host type=license count=50  
sacctmgr add resource name=matlab server=rlm_host cluster=oldone allowed=70  
sacctmgr add resource name=matlab server=rlm_host cluster=newone allowed=30
```

To reiterate, “server” in this resource declaration is informational only—even if the servers are on `rlm_host`, or whether it is `flexlm_1`, neither of these are integrating or binding.

They are, however, new requirements in specifying how to request the resource:

```
sbatch -L matlab@rlm_host my_script.sh
```

This is in contrast to the local license that looked like:

```
sbatch -L matlab my_script.sh
```

License Types

Remote Licenses “server”

But this granularity would permit multiple licenses commitments to be handled independently:

```
sacctmgr add resource name=matlabA server=college_math type=license count=50
sacctmgr add resource name=matlabA server=college_math cluster=oldone allowed=30
sacctmgr add resource name=matlabA server=college_math cluster=newone allowed=70
```

```
sacctmgr add resource name=matlabB server=college_eng type=license count=450
sacctmgr add resource name=matlabB server=college_eng cluster=oldone allowed=50
sacctmgr add resource name=matlabB server=college_eng cluster=newone allowed=450
```

And since these are saved in the database (as all remote licenses are managed), there is now an accurate accounting of the license usage, which one and on which cluster.

```
sbatch -L matlabA@college_math my_script.sh
```

License Types

Trackable Resources (TRES)

And to top it off, licenses are TRES!

```
sacctmgr modify account where name=engineering set GrpTRES=license/matlabB=350  
sacctmgr modify account where name=math set GrpTRES=license/matlabA=40
```

By setting limits on these specific accounts, all other accounts might still have a chance of using them when in high demand, e.g., 100 matlabB licenses might be allocatable to other non-named groups—or 450 might be allocatable!

Simply put, we are given the full precision of putting limits/caps on the usage, depending on the demands of our licensing restrictions and site policy.

License Types

Trackable Resources (TRES)

So all the other ways in which you might modify the behavior of a TRES apply in addition to QOS, such as in partitions and billing weights:

```
TRESBillingWeights=CPU=1.0,gres/gpu=3.0,license/matlabA=4.0,license/matlabB=4.5
```

License Types

Thank you!

Any questions?